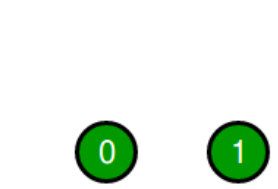
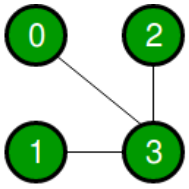
VERETX COVER PROBLEM

Problem –Given a graph G (V, E) and a positive integer k, the problem is to find whether there is a subset V' of vertices of size at most k, such that every edge in the graph is connected to some vertex in V'.

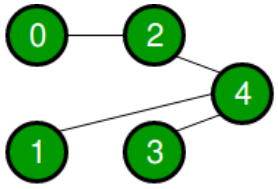
- ✓ A Vertex cover of an undirected graph is a subset of its vertices that for edge (u, v) of the graph, either 'u' or 'v' is in vertex cover.
- ✓ The set covers all edges of the given graph.
- ✓ Given an undirected graph, the vertex cover problem is to find minimum size vertex cover.



Minimum vertex cover is empty{}



Minimum vertex cover is {3}



Minimum vertex cover is {4, 2} or {4, 0}

G(V,E)

V={0,1,2,3}

E= {<0,3>,<1,3>,<2,3>}

V'= {3}

V={0,1,2,3,4}

E={<0,2>,<2,4>,<1,4>,<3,4>}

V'={4,0}

VERETX COVER PROBLEM-NP-COMPLETE

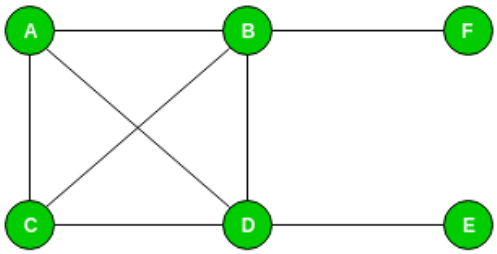
I. Proof that vertex cover is in NP

- ✓ If any problem is in NP, then, given a 'certificate' (a solution) to the problem and an instance of the problem (a graph G and a positive integer k, in this case), we will be able to verify (check whether the solution given is correct or not) the certificate in polynomial time.
- ✓ The certificate for the vertex cover problem is a subset V' of V, which contains the vertices in the vertex cover. We can check whether the set V' is a vertex cover of size k.
- ✓ It is plain to see that this can be done in polynomial time.
- ✓ Thus the vertex cover problem is in the class NP.

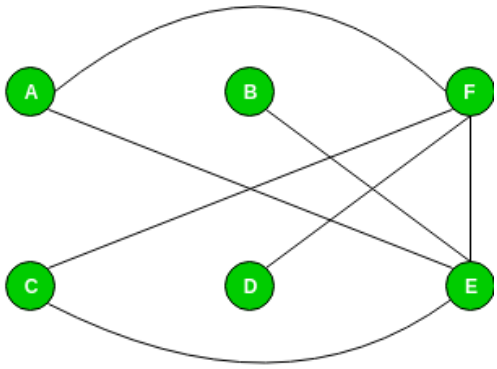
II. Proof that vertex cover is NP Hard

- ✓ To prove that Vertex Cover is NP Hard, we take some problem which has already been proven to be NP Hard, and show that this problem can be reduced to the Vertex Cover problem.
- ✓ For this, we consider the Clique problem, which is NP Complete (and hence NP Hard).
    - We need to show that any instance (G, k) of the Clique problem can be reduced to an instance of the vertex cover problem.
    - Consider the graph G' which consists of all edges not in G, but in the complete graph using all vertices in G.
    - Let us call this the complement of G.

- Now, the problem of finding whether a clique of size k exists in the graph G is the same as the problem of finding whether there is a vertex cover of size |V| –k in G'.
- we can say that there is a clique of size k in graph G if and only if there is a vertex cover of size |V| –k in G', and hence, any instance of the clique problem can be reduced to an instance of the vertex cover problem.
- Thus, vertex cover is NP Hard.
- Since vertex cover is in both NP and NP Hard classes, it is NP Complete.



G
V' ={A, B, C, D}



G'
V'' ={E, F}

TSP-HAMILTONIAN CYCLE

KNAPSACK-SUBSET-SUM

CLIQUE-BOOLEAN SATISFIABILITY PROBLEM

VERTEX COVER-CLIQUE